Fiscal Year:	FY 2008	**Task Last Updated:**	FY 08/12/2010
PI Name:	Pollock, Neal Ph.D.		
Project Title:	Mechanisms of Musculoskeletal-Induced Nucleation in Altitude Decompression Stress		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
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Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2003 Biomedical Research & Countermeasures 03-OBPR-04
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No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
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Flight Program:			
Flight Assignment:	NOTE: project extended until 6/30/2011 per JSC (06/2010)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Gernhardt, Michael (NASA Johnson Space Center) Vann, Richard (Duke University) Conkin, Johnny (Baylor College of Medicine)		
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Task Description:	INTRODUCTION: Musculoskeletal activity has the potential to both improve and compromise decompression safety, depending on the intensity, sequence and level of tissue supersaturation. Exercise enhances inert gas elimination during oxygen breathing prior to decompression and it may also promote bubble nuclei formation, which can lead to gas phase separation and growth resulting in increased DCS risk. The timing, sequence and intensity of musculoskeletal activity may be critical to the net effect. There are limited data available to evaluate the cost:benefit relationship. Understanding the relationship is important to improve our scientific understanding of the underlying mechanisms of nucleation in exercise prebreathe protocols and to quantify variable risk in gravity and microgravity environments when musculoskeletal effort can differ substantially. Data gathered during prebreathe reduction program (PRP) studies combined multiple variables (prebreathe exercise and microgravity simulation) to produce a procedure now used by astronauts preparing for extravehicular activity (the Phase II protocol). The PRP results would serve as control data for this NASA/Duke multi-center study to investigate the influence of individual variables (exercise and ambulation) on bubble formation and the subsequent risk of decompression sickness. HYPOTHESIS: 1) Ambulation at altitude, under supersaturated conditions, generates bubble nuclei, resulting in increased decompression stress. 2) Preflight ambulation during undersaturated conditions does not significantly increase nucleation and decompression stress on the exercise prebreathe protocols. 3) Light exercise following heavy exercise facilitates the removal of heavy exercise induced bubble nuclei, resulting in lower decompression stress. METHODS: Four separate experiments would replicate the PRP Phase II protocol, each with a different exception. Experiment I – Subjects would be ambulatory both preflight and at altitude instead of non-ambulatory throughout. A minimum of 25 and a maximum of 50 subjects would be tested. A Fisher exact test would be used to compare the results of the test and control groups. The protocol would be stopped, when the DCS or VGE observations rise to the level of statistical significance (3/25, 5/50 DCS, $p<0.05$) Experiment II - Subjects would remain non-ambulatory during the preflight period and ambulatory at altitude. Experiment II would use the same statistical design, as experiment I. Experiment III – Subjects would be ambulatory during the preflight period and non-ambulatory at altitude. Experiment IV - The order of heavy and light exercise would be reversed, with the light exercise occurring first. EXPECTED RESULTS. It is anticipated these experiments will show that for exercise prebreathe protocols, decompression stress is related to musculoskeletal nucleation, and that nucleation can be controlled to reduce decompression stress. Specifically, that ambulation during the altitude exposure results in increased decompression stress, but that ambulation prior to altitude exposure does not. Additionally, the results will show that reversing the order of light and heavy exercise will result in higher decompression stress, suggesting that light exercise facilitates the removal of heavy exercise induced nucleation. These results will demonstrate that the degree, sequence and level of supersaturation under which musculoskeletal activity is performed are important in controlling nuclei generation and subsequent decompression stress.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	[Ed. note=information received June 2010 for 5/9/2008 report] Completion of the first four phases of the PRP program yielded a successful two hour prebreathe protocol that combined an initial 10 minute period of heavy exercise on a cycle ergometer, followed by light intermittent exercise, followed by a final 40 minutes of resting prebreathe. Laboratory simulations of this protocol (Phase II) were clearly superior to the others tested, with no DCS in 45 subjectexposures. The protocol was first implemented operationally during STS-104 in July 2001. A total of 34 spacewalks employed this protocol, the last pair on April 08, 2003, with no complications. The limitation of the existing “out-of-suit” exercise protocol is its reliance on a complicated infrastructure that includes the possibility of up to 21 single-point failures, and high oxygen usage associated with the open-circuit oxygen delivery system. The objective of Phase V was to investigate the viability of establishing an equally effective prebreathe protocol that could be conducted completely in the closed-circuit space suit, with the result of limiting the failure modes and significantly reducing the demand on oxygen consumables. Phase V trials were conducted between November 2002 and March 2008. Exposures were completed for Protocol V-1 11/25/02-1/22/03; V-2 3/4/03-3/27/03; V-3 5/12/03-12/16/03; V-4 8/17/04-9/15/05; and V-5 10/13/06-3/28/08. One of the elements evaluated in the Phase V protocols was the use of intermittent exercise, an effort driven by both practical and theoretical considerations. Exercise in the suit can generate local hot spots and in a pilot study astronauts preferred brief periods of exercise followed by rest rather than a continuous 10 minute exercise period. Additionally, there is an inherent asymmetry between the rapid onset of blood flow in response to exercise and the gradual relaxation of blood flow following the cessation of exercise. Theoretically, the use of intermittent exercise should result in more cumulative blood flow and nitrogen elimination for a given amount of total exercise, achieving a better balance between inert gas elimination and micronuclei generation.
Bibliography Type:	Description: (Last Updated: 03/27/2017)
Articles in Peer-reviewed Journals	Conkin J, Pollock NW, Natoli MJ, Martina SD, Wessel JH 3rd, Gernhardt ML. "Venous gas emboli and ambulation at 4.3 psia." Aerosp Med Hum Perform. 2017 Apr;88(4):370-6. https://dx.doi.org/10.3357/AMHP.4733.2017 ; http://www.ingentaconnect.com/content/asma/amhp/2017/00000088/00000004/art00003 , Apr-2017